

XMLHttpRequest “Ping” Sweeping in Firefox 3.5+ ha.ckers.org web application security lab

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Content

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XMLHttpRequest “Ping” Sweeping in Firefox 3.5+

[Jeremiah](#) brought my attention to the new Firefox 3.5+ [CORS \(Cross-Origin Resource Sharing\)](#) which is a way to do a cross domain XMLHttpRequest. Does that sound scary? Well, it is, but there’s been a ton of work into hardening it. It has all sorts of cross domain opt-in verification built into it to limit the abuse. Honestly, if you look at [the people](#) who were acknowledged in it’s construction, it’s a who’s who of people who understand cross domain browser security issues. So it wasn’t surprising that it was fairly free of obvious flaws.

Anyway, I was poking around with it and I noticed that it had one fairly strange issue. Although an attacker is not allowed to know if the page was there or not (only if it was allowed to see the content or not), the attacker is still allowed to make an initial request. In doing so that initial request can be used as a pseudo “ping” sweep. You can tell if the site is there or not because it will either return immediately (latency and threading applies) or it will wait around much longer (between 20-75 seconds on the several networks I’ve run this on) before the browser gives up. That timing difference is pretty substantial - and as a result you can enumerate a substantial amount of internal address space behind the victim’s firewall and relatively quickly. I created [a demo here](#) (works only in Firefox 3.5+ and you must enable JavaScript *globally* for this to work). It won’t work if you just whitelist ha.ckers.org you have

to globally allow JavaScript if you use Noscript for the demo to work - and you must disable ABE in Noscript as well.

You can read the page for the details, like the fact that basic and digest authentication popups are suppressed which makes this technique ideal for Intranets where those are common and would normally alert a user to the fact that something was wrong in the browser. It also doesn't matter whether you do or don't have port 80 open for this to work, I should note that there is a IE8.0 version of Firefox's XMLHttpRequest called [XDomainRequest](#), but I didn't have much time this weekend to try to get it working in both browsers so I have no idea if it has the same issue or not.

Incidentally, Jeremiah and I both gave the thumbs up to the idea of a cross domain XHR several years ago when the Mozilla team first asked us about the concept. Because there are so many other things wrong with the browser Jeremiah and I told them that it wouldn't change much - the browser is already so broken from a security perspective that it really didn't matter - a sad commentary thinking back. Of course, it really is all about the implementation.

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